Jump Modelling with Dirichlet Process Mixture

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Motivation

Jump Diffusion Models

Jump diffusion models are introduced to:

- Model the sudden and extreme price changes that is not captured by volatility models,
- Capture the fat tails of return distribution,
- Early literature on jump models: Merton (1976), Bates (1996), Kou (2002), Kou (2007)
- Adding jumps to stochastic volatility (SV) models better explain the empirical implied volatility (IV) smile (Pan, 2002; Eraker *et al.*, 2003; Eraker, 2004).
- Hence provide better specification for option pricing.

Classic SVJ model:

Let p_t be the logarithm of the asset price P_t at time t and V_t be the latent variance.

$$\begin{aligned} dp_t &= \mu_t dt + (V_t)^{1/2} dW_t^p + Z_t dN_t, \\ d(\log(V_t)) &= \kappa(\theta - \log(V_t))dt + \gamma dW_t^v, \end{aligned} \tag{1}$$

where dN_t represents the jump occurrence and Z_t is the jump size, which is assumed to follow a particular distribution.

Some further investigation of jump structure: dynamic jumps (Maheu and McCurdy, 2004); jump clustering (Aït-Sahalia *et al.*, 2015); jump directions (Maneesoonthorn *et al.*, 2017).

Simulation Example: Mixed jump density

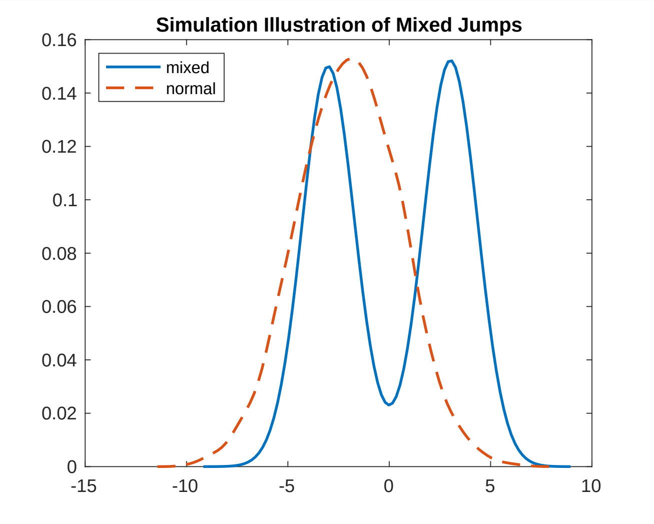

Figure 1: Simulation illustration for the motivation of proposing mixed jumps. The blue solid line shows the mixed normal density and the red dashed line is a single normal density.

Motivation

Parametric SVJ model:

- Assuming a particular distribution for jump size, usually normal or exponential distribution;
- Could result higher mass around 0 for jump size density;
- The misspecification in jump distribution will be reflected in less accurate option prices and IV.

Motivation

Our proposed DPM jump model aims to:

- Model the jump size distribution flexibly and more data driven;
- Recover the empirical jump size distribution;
- *Provide better model specification for option pricing and IV forecasting.*

Model Specification

Discrete time static SV-DPM jump model

On day t, let r_t be the log return and ,h_t be the log-variance of return and ΔN_t represents the jump arrival and Z_t is the jump size.

$$\begin{aligned}
r_t &= \exp(h_t/2)\varepsilon_t^r + \Delta N_t Z_t, \\
h_t &= \phi_0 + \phi_1 h_{t-1} + \sigma_t^h \varepsilon_t^h, \\
\Delta N_t &\sim \text{Bernoulli}(\delta), \\
Z_t &\sim \sum_{k=1}^{\infty} w_k N(\mu_k, \sigma_k^2), \\
w_k &= \nu_k \prod_{l=1}^{k-1}(1-\nu_l), \text{ with } \nu_k \sim \text{Beta}(1, \alpha),
\end{aligned} \tag{2}$$

where $\varepsilon_t^r, \varepsilon_t^{BV}, \varepsilon_t^h$ follow $N(0,1)$. We model jump size Z_t distribution as a stick-breaking process (SBP) (Sethuraman, 1994).

Further development: Practical motivation

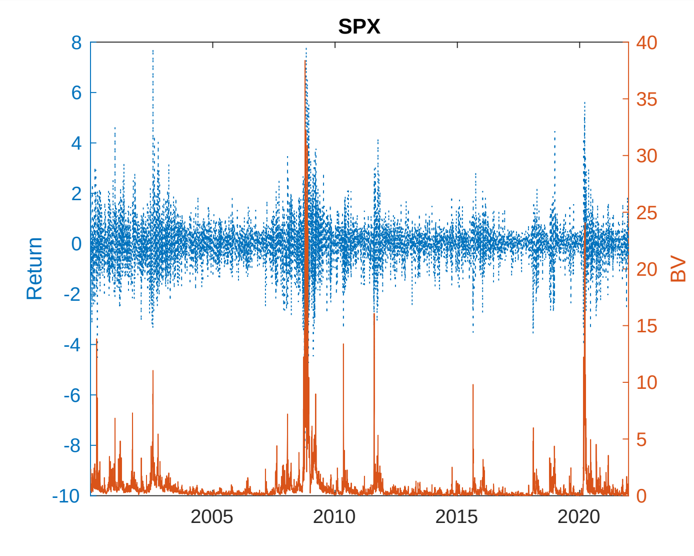

Figure 2: Empirical return and bipower variation for SPX.

Discrete time dynamic SV-DPM jump model

Incorporating bipower variation:

$$BV_t = \frac{\pi}{2}\left(\frac{N}{N-1}\right)\sum_{i=2}^{N}|r_{t_i}||r_{t_{i+1}}| \xrightarrow{p} IV_t, \text{ as } N \to \infty, \tag{3}$$

where $r_{t_i} = p_{t_{i+1}} - p_{t_i}$ is the ith of N observed returns between trading day t and $t+1$ (Barndorff-Nielsen and Shephard, 2004).

The specification of jump size becomes:

$$\begin{aligned} Z_t &\sim \sum_{k=1}^{\infty} w_k N(\mu_k (B\bar{V}_t)^{1/2}, \sigma_k^2), \\ w_k &= v_k \prod_{l=1}^{k-1}(1-v_l), \text{ with } v_k \sim \text{Beta}(1,\alpha), \\ (B\bar{V}_t)^{1/2} &= \left[\frac{1}{n}\sum_{m=1}^{n}(BV_{t-m})\right]^{1/2} \end{aligned} \tag{4}$$

Inference

Joint Posterior

Let $\Theta = \{\phi_0, \phi_1, \sigma_h^2, \sigma_{BV}^2\}$ and $\mathbf{\Phi} = \{\boldsymbol{w}, \mu, \boldsymbol{\sigma^2}, \delta\}$ where $\mu = \{\mu_1, \ldots, \mu_\infty\}$ and $\boldsymbol{\sigma^2} = \{\sigma_1^2, \ldots, \sigma_\infty^2\}$.

For dynamic DPM model:

$$
\begin{aligned}
P(\mathbf{\Theta}, \mathbf{\Phi}, \boldsymbol{h}, \boldsymbol{\Delta N}, \boldsymbol{Z} | \boldsymbol{r}, \boldsymbol{(B\bar{V})}^{1/2}) \propto\ & P(\phi_0, \phi_1 | \sigma_h^2) P(\sigma_h^2) P(\delta) P(h_0) \\
& \prod_{t=1}^{T} \Big[P(\Delta N_t | \delta) P(h_t | h_{t-1}, \phi_0, \phi_1, \sigma_h^2) P(r_t | h_t, Z_t, \Delta N_t) \\
& \prod_{k=1}^{\infty} P(\mu_k, \sigma_k^2) P(w_k) P(Z_t | \mu_k (B\bar{V}_t)^{1/2}, \sigma_k^2) \Big]
\end{aligned}
\tag{5}
$$

Slice Sampler

An auxiliary variable u_t is introduced who is uniformly distributed conditional on $\boldsymbol{w}$ and indicator variable of jump mixture S_t^z.

$$u_t | S_t^z, \boldsymbol{w} \sim U(0, w_{S_t^z}). \tag{6}$$

The joint distribution of u_t and Z_t is:

$$\begin{aligned} p(u_t, Z_t | \cdot) &= \sum_{k=1}^{\infty} w_k U(u_t | 0, w_k) N(Z_t | \mu_k, \sigma_k^2), \\ &= \sum_{k=1}^{\infty} I(u_t < w_k) N(Z_t | \mu_k, \sigma_k^2), \\ &= \sum_{k \in A_t} N(Z_t | \mu_k, \sigma_k^2) \end{aligned} \tag{7}$$

where $A_t = \{k : u_t < w_k\}$. Notably, A_t here is a set with finite elements (Neal, 2003; Walker, 2007).

MCMC Algorithm

With total M iterations, the MCMC algorithm is:

Algorithm 1 Full Gibbs sampling algorithm

Require: prior, $\boldsymbol{r}$

Initialize $(i=0)$ $\boldsymbol{\Theta}, \boldsymbol{S}$ for initialization of $\boldsymbol{h}$;

Initialize $(i=0)$ $(\mu_{S_t^z}, \sigma^2_{S_t^z}), \boldsymbol{S}^z$ for initialization of $\boldsymbol{Z}$.

Initialize $(i=0)$ δ

For each MCMC iteration i

1. Sample $\boldsymbol{\Delta N}^{(i)} | \boldsymbol{h}^{(i-1)}, \boldsymbol{r}, \boldsymbol{Z}^{(i-1)}, \delta^{(i-1)}$
2. Sample $\delta^{(i)} | \boldsymbol{\Delta N}^{(i)}$
3. Sample $\boldsymbol{w}^{(i)} | \boldsymbol{S}_z^{(i-1)}, \boldsymbol{u}^{(i)} | \boldsymbol{w}^{(i)}, \boldsymbol{S}_z^{(i-1)}$
4. Sample $(\mu_{S_t^z}, \sigma^2_{S_t^z})^{(i)} | \boldsymbol{S}_z^{(i-1)}, \boldsymbol{Z}^{(i-1)}$
5. Sample $\boldsymbol{S}_z^{(i)} | \boldsymbol{Z}^{(i-1)}, (\mu_{S_t^z}, \sigma^2_{S_t^z})^{(i)}, \boldsymbol{w}^{(i)}, \boldsymbol{u}^{(i)}$
6. Sample $\boldsymbol{Z}^{(i)} | \boldsymbol{h}^{(i-1)}, \boldsymbol{r}, \boldsymbol{\Delta N}^{(i)}, \boldsymbol{S}_z^{(i)}, (\mu_{S_t^z}, \sigma^2_{S_t^z})^{(i)}$
7. Sample $\boldsymbol{h}^{(i)} | \boldsymbol{r}, \boldsymbol{\Delta N}^{(i)}, \boldsymbol{Z}^{(i)}, \boldsymbol{S}^{(i-1)}, \boldsymbol{\Theta}^{(i-1)}$ ▷ using 7-component mixture (Kim *et al.*, 1998).
8. Sample $\boldsymbol{\Theta}^{(i)} | \boldsymbol{h}^{(i)}$
9. Sample $\boldsymbol{S}^{(i)} | \boldsymbol{h}^{(i)}, \boldsymbol{r}$ ▷ $\boldsymbol{S}$ is the indicator for seven point mixture.
10. Go to step (1) and repeat for M iterations.

return $\boldsymbol{h}, \boldsymbol{Z}, \boldsymbol{\Delta N}, \boldsymbol{\Theta}$

Simulation: Model Sensitivity

Simulation Analysis

The purpose of this simulation is to investigate the model performance:

- when true jump intensity changes;
- when true jump size changes;
- how our model performs compared with parametric SVJ model where jump size $Z_t \sim N(\mu_z, \sigma_z^2)$.

True Data Generating Process

We generate $T = 5000$ observations from the following true DGP:

$$\begin{aligned}
r_t &= \exp(h_t/2)\varepsilon_t^r + \Delta N_t Z_t, \\
h_t &= -0.06 + 0.94 h_{t-1} + 0.38\varepsilon_t^h, \\
\log(BV_t) &= h_t + 0.4\varepsilon_t^{BV}, \\
B\bar{V}_t &= \frac{1}{5}\sum_{m=1}^{5} BV_{t-m}, \\
\Delta N_t &\sim \text{Bernoulli}(\delta), \\
Z_t &\sim 0.6 \times N(-\mu_z e^{h_t/2}, 0.8) + 0.4 \times N(\mu_z e^{h_t/2}, 0.6).
\end{aligned} \tag{8}$$

We compare the model performance on a various range of $\delta = \{1\%, 5\%, 10\% 15\%\}$ and μ_z = {2, 2.5, 3, 3.5} away from the conditional mean of return distribution.

Simulation Results: ROC

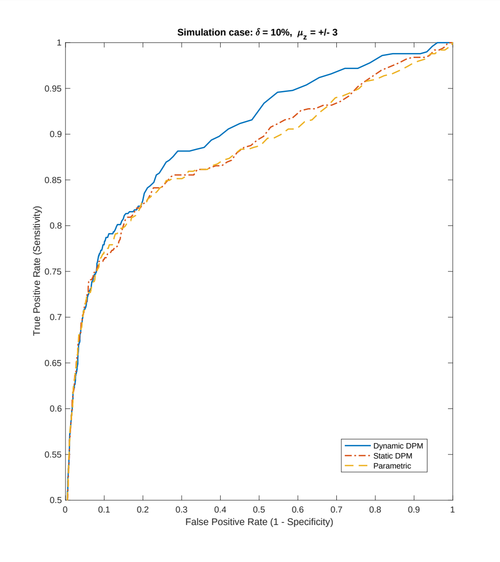

Figure 3: ROC for one particular simulation case where the true DGP with jump intensity $\delta = 10\%$ and the mean of jump size $\mu_z = +/-3$.

Simulation Results: AUC

Area under the ROC curve (AUC)			
	Dynamic DPM	Static DPM	Parametric SVJ
A: Jump Intensity Analysis ($\mu_z = 3$)			
$\delta = 1\%$	0.8960	0.8791	**0.9043**
$\delta = 5\%$	**0.9054**	0.8883	0.8819
$\delta = 10\%$	**0.9004**	0.8816	0.8788
$\delta = 15\%$	**0.9208**	0.9018	0.9018
B: Jump Size Analysis ($\delta = 10\%$)			
$\mu_z = 2$	0.7966	0.7932	**0.7974**
$\mu_z = 2.5$	**0.8491**	0.8270	0.8287
$\mu_z = 3$	**0.9004**	0.8816	0.8788
$\mu_z = 3.5$	**0.9363**	0.9160	0.9171

Table 1: Area under the ROC curve (AUC) for both jump intensity analysis and jump size analysis under simulation. The larger AUC, the more accurate the model is in terms of detecting true jumps. The bold numbers are the maximum of each cases.

Marginal Density of Jump Size

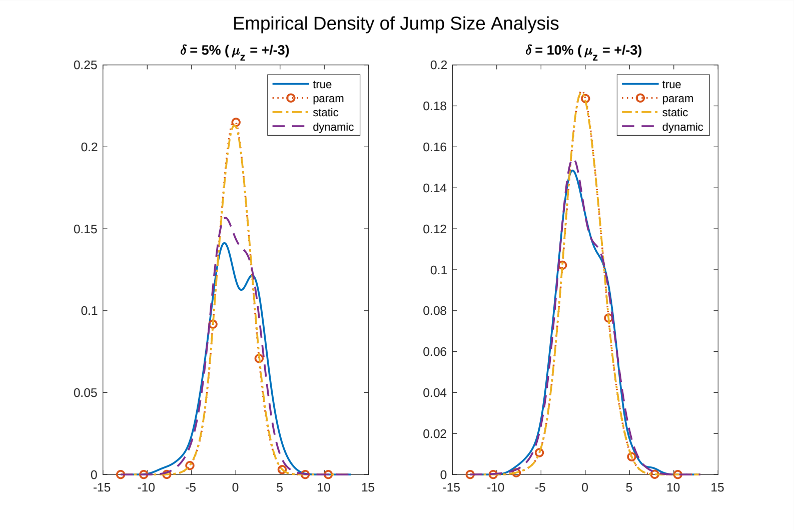

Figure 4: The **estimated jump kernel** for DPM jump model under **different true DGPs**. The blue dashed line represents the true jump density. The red, yellow and purple lines represent parametric SVJ, static DPM and dynamic DPM, respectively.

Preliminary Empirical Analysis

Data

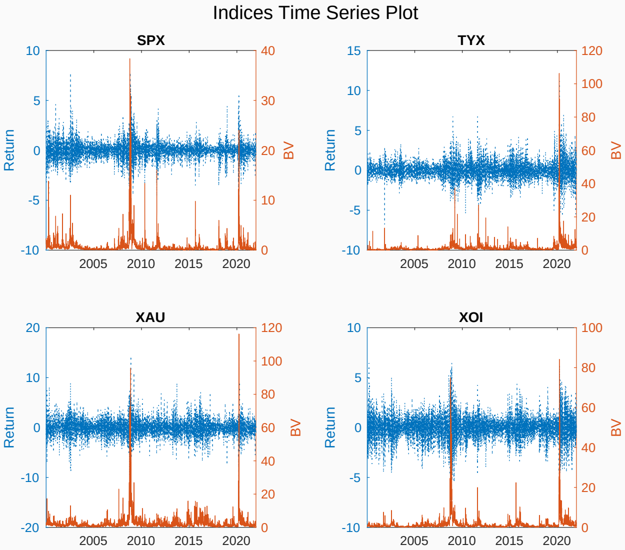

Figure 5: Time series plots of four financial indices. SPX - S&P 500 index, TYX - Treasury yield index (30 year bond), XAU - gold and silver index and XOI - oil index.

Estimated jump size distribution

Estimated jump size distributions for financial indices		
Parameters	Posterior Mean/Mode (for K)	90% Posterior Interval/Range (for K)
	SPX	
K	2	[2,10]
μ_z	1.2220; -1.5656	(0.9236, 1.5885); (-1.9709, -1.1900)
σ_z	0.4022; 0.4787	(0.3192,0.5044); (0.3703, 0.5934)
	TYX	
K	2	[2,10]
μ_z	-1.2943; 1.5057	(-1.8216, -0.8522); (1.2094, 1.8179)
σ_z	0.5380; 0.4849	(0.3946,0.7217); (0.3668, 0.6322)
	XAU	
K	2	[2,10]
μ_z	2.0420; -1.6369	(1.6303. 2.4210); (-1.9333,-1.3232)
σ_z	0.5586; 0.6294	(0.4038, 0.7638); (0.4344, 0.8770)
	XOI	
K	2	[2,6]
μ_z	-1.7014; 1.1320	(-2.0487, -1.3210); (0.8083, 1.4869)
σ_z	0.5556; 0.5728	(0.4019, 0.7758); (0.4238,0.7506)

Table 2: Estimated jump component K, and the corresponding mean and variance of jump size μ_z, σ_z. We report the posterior mean/mode (for K) and 90% posterior interval or range for K.

Results I: Estimated jump intensity

Estimated Jump Intensity for Financial Indices				
	SPX	TYX	XAU	XOI
Parametric	0.0975	0.1482	0.1900	0.1329
	(0.0661, 0.1310)	(0.1054, 0.1965)	(0.1533 ,0.2345)	(0.0955, 0.1756)
DPM	0.1071	0.1107	0.1341	0.1249
	(0.0757, 0.1435)	(0.0744, 0.1538)	(0.1031, 0.1653)	(0.0876, 0.1645)

Table 3: Post-convergence **posterior mean of jump intensity** δ and their 90% posterior interval.

Estimated Jump Density

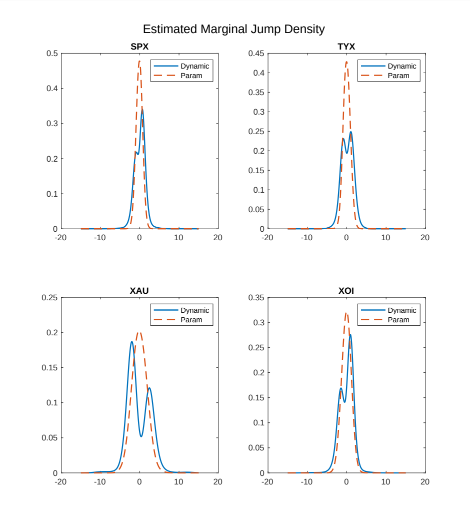

Figure 6: The **estimated marginal jump density** of both parametric and DPM jump models for four financial indices. The blue solid line represents the jump kernel estimated from **dynamic** DPM jump model. The red dashed line represents the estimated jump kernel from parametric jump model.

Jump Size Density Over Time

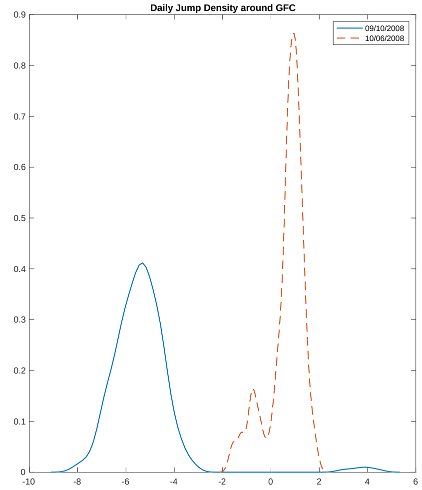

Figure 7: Jump density on two particular days during Global Financial Crisis period.

Predictive Density of Returns

We produce real time one step ahead predictive density using a rolling window method, and evaluate the predictive density of returns using the strictly proper scoring rules:

- Log score (LS)
- Censored likelihood score (CLS) at 10% and 90% quantile

Prediction Evaluation

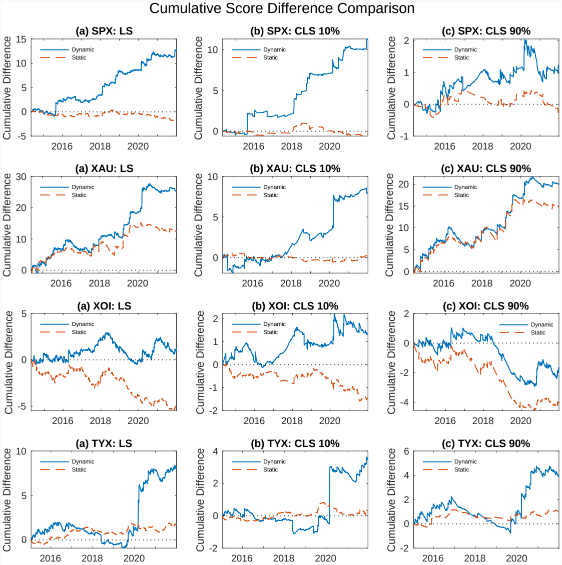

Figure 8: Score evaluation of predictive density for all indices.

Empirical application in IV smile

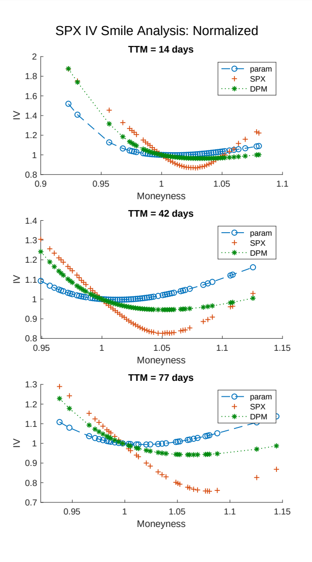

Figure 9: Normalized IV comparison for SPX index.

Summary

The model we proposed here provides:

- Dynamic DPM provides better estimation results than parametric SVJ and static DPM;
- Recover the empirical jump size distribution driven by data;
- Provides better predictive return density than parametric SVJ model, especially during volatile periods.

The model we implemented here considers relatively simple structure of jumps:

- Jumps are assumed to be i.i.d,
- Volatility jumps and leverage effects are missing

Questions?

References

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Appendix

Implied Volatility: Definition

Implied volatility is the volatility that is implied by the market option price. Let c_t^m be market call option price and solve for $\sigma = \hat{\sigma}$ such that

$$c_t^m = c_t^{BS}(\sigma), \tag{9}$$

where c_t^{BS} is the call option price calculated using Black-Scholes model (Black and Scholes, 1973).

Models

Assume the stock price follows the structure as below:

1. SV model
2. Parametric SVJ model
3. SV model with mixture jumps

Let X be strike price, r_f be risk-free rate and T_m be time-to-maturity. The call option price can be simulated via Monte Carlo simulation as:

$$C(S_T, X, T_m) = \frac{1}{(1+r_f)^{T_m}} E^Q[\max(S_{T_m} - X, 0)]. \tag{10}$$

DAGs with Slice Sampler

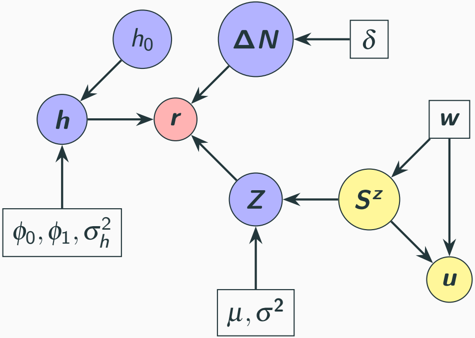

Figure 10: DAGs of the DPM jump model structure. Red circles represents observed data, purple circles are latent variables. Parameters are represented as rectangular. Yellow circles represents the auxiliary variables where $\boldsymbol{u}$ is introduced for slice sampler and $\boldsymbol{S}^z$ is an indicator functions for components.